

JTMK Project Report:

Mobile-Based Campus Navigation App Using QR Codes

Project Title:

Smart Campus Navigation System for New Students Using QR Code Technology

Team Members:

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Introduction:

Navigating a large campus can be overwhelming for new students. This project introduces a mobile-based navigation system that utilizes QR codes placed around the campus. Scanning a code leads users to a mobile-friendly interface displaying directions, room details, and building info — enhancing the student onboarding experience at JTMK.

Objectives:

1. To design a QR-based navigation system for campus wayfinding.
2. To develop a mobile web interface for dynamic location access.
3. To simplify navigation for new students and visitors.
4. To reduce reliance on printed maps and physical signage.

Tools and Technologies Used:

- Hardware: QR Code Stickers, Smartphones
- Software: HTML5, JavaScript, jQuery Mobile
- Tools: QR Code Generator, Google Maps API
- Languages: HTML/CSS, JavaScript

Features:

- Location-based landing pages via QR scan
- Interactive campus map integration
- Directions to labs, offices, lecture halls
- Mobile-friendly and lightweight interface

Implementation Process:

1. Mapped key locations in the JTMK building.
2. Generated unique QR codes for each point of interest.
3. Built mobile pages using jQuery Mobile framework.
4. Linked QR codes to their corresponding HTML pages.
5. Conducted pilot test with first-semester students.

Results:

- Users reported faster orientation during first week.
- 100% mobile compatibility with both iOS and Android.
- Reduced need for printed guides by 85%.

Challenges:

- Some QR codes became damaged or faded.
- Users needed internet access for full functionality.
- Mapping small internal rooms was difficult.

Future Enhancements:

- Add AR (augmented reality) features for visual directions.
- Implement offline mode using PWA (Progressive Web App).
- Translate pages into multiple languages for international students.

Conclusion:

The project provides an efficient and scalable way for JTMK to guide students around campus. It enhances digital literacy and supports a smart campus vision aligned with modern educational needs.

References:

1. QR Code Basics – <https://www.qr-code-generator.com>
2. jQuery Mobile Docs – <https://jquerymobile.com/demos/1.4.5/>
3. Google Maps JavaScript API – <https://developers.google.com/maps/documentation/javascript>